NAME: _____ Geometry (FL B.E.S.T.)

5.4 Triangle Proportionality

The **Triangle Proportionality Theorem** (Side-Splitter Theorem) says: if a line parallel to one side of a triangle intersects the other two sides, it divides those sides proportionally. For $\overline{DE} \parallel \overline{BC}$ with D on $\overline{AB}$ and E on $\overline{AC}$, this gives $\frac{AD}{DB} = \frac{AE}{EC}$, or equivalently the "whole-to-part" form $\frac{AD}{AB} = \frac{AE}{AC}$. The **converse** is also true: if a line divides two sides proportionally, it must be parallel to the third side. A useful corollary extends this to **three or more parallel lines**: they cut any two transversals into proportional segments. A special case is the **Midsegment Theorem**: the segment joining the midpoints of two sides is parallel to the third side and exactly half its length. This worksheet applies these ideas numerically, algebraically, and with a coordinate proof.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. $\overline{DE} \parallel \overline{BC}, AD = 6, DB = 4, AE = 9$. Find EC.

2. $\overline{DE} \parallel \overline{BC}, AD = 5, AB = 12, AE = 10$. Find AC.

3. $AD = 8, DB = 4, AE = 10, EC = 5$. Is $\overline{DE} \parallel \overline{BC}$?

4. $AD = 2, DB = 3, AE = 4, EC = 5$. Parallel?

5. $\overline{DE} \parallel \overline{BC}, AD = x, DB = 10, AE = 6, EC = 15$. Find x.

6. Third side is 22. Find the midsegment length.

7. $\overline{DE} \parallel \overline{BC}, AD = 4, DB = 8, AC = 18$. Find AE.

8. $\overline{FG} \parallel \overline{BC}, AF = 3, FB = 9, AG = 2$. Find GC.

9. Midsegment length is 7. Find the third side.

10. $AD = 7, AB = 21, AE = 5$. If $\overline{DE} \parallel \overline{BC}$, find AC.

11. $AD = 9, DB = 6, AE = 12, EC = 8$. Parallel?

12. $\overline{DE} \parallel \overline{BC}, AD = 10, DB = 15, AE = 8$. Find EC.

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13. Three parallel lines cut two transversals into segments $4, 6$ on the first, and $6, x$ on the second (matched in order). Find x.

14. Triangle has vertices $A(0,0), B(10,0), C(4,8)$. Find midpoints M (of $\overline{AB}$) and N (of $\overline{AC}$), the length MN, and verify $\overline{MN} \parallel \overline{BC}$ with $MN = \frac{1}{2}BC$.

15. $\overline{DE} \parallel \overline{BC}$, $AD = 2x + 1$, $DB = x + 5$, $AE = 12$, $EC = 8$. Find x, then AD, DB, and AB.

✓ Answer Key

1. 6
2. 24
3. Yes; both ratios = 2
4. No; $\frac{2}{3} \neq \frac{4}{5}$
5. 4
6. 11
7. 6
8. 6
9. 14
10. 15
11. Yes; both ratios = $\frac{3}{2}$
12. 12
13. $x = 9$
14. $M(5,0)$, $N(2,4)$; $MN = 5$, $BC = 10$; both have slope $-\frac{4}{3}$, and $MN = \frac{1}{2}BC$
15. $x = 13$; $AD = 27$, $DB = 18$, $AB = 45$